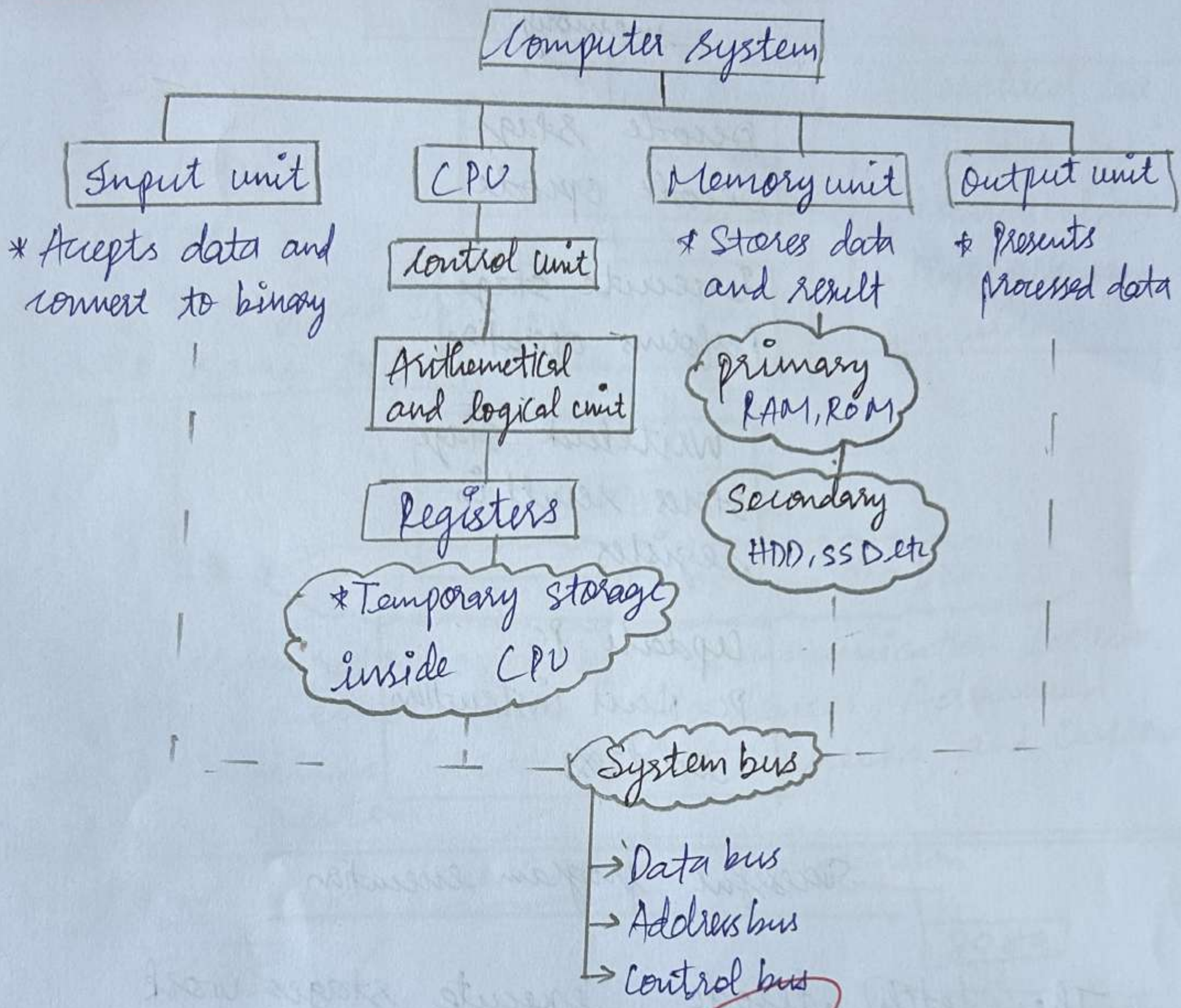
1. Interaction of functional units in a computer system



All unit works together through System bus to execute instruction efficiently

2. Instruction Execution cycle

It is step by step process used by the CPU to execute a program

Fetch Stage
Fetch instruction from memory

↓

Decode Stage
Decode opcode

↓

Execute stage
Performs operation

↓

Writeback stage
Stores result in register

↓

Update PC
PC = Next instruction address

↓

Successful program execution

The ~~fetch~~, decode, execute stages work together in sequence to ensure accurate, efficient and continuous program execution.

Bus organization in Computer Architecture

Bus Organization

Single bus
- → One bus for all transfers
- → CPU, Memory and I/O share same bus

Multi bus
- → seperate bus for data, address and control
- → More than one path

Hierarchical bus
- → Multiple bus in hierarchical levels
- → high performance, scalable

Connects

CPU ⟷ MEMORY ⟷ I/O DEVICES

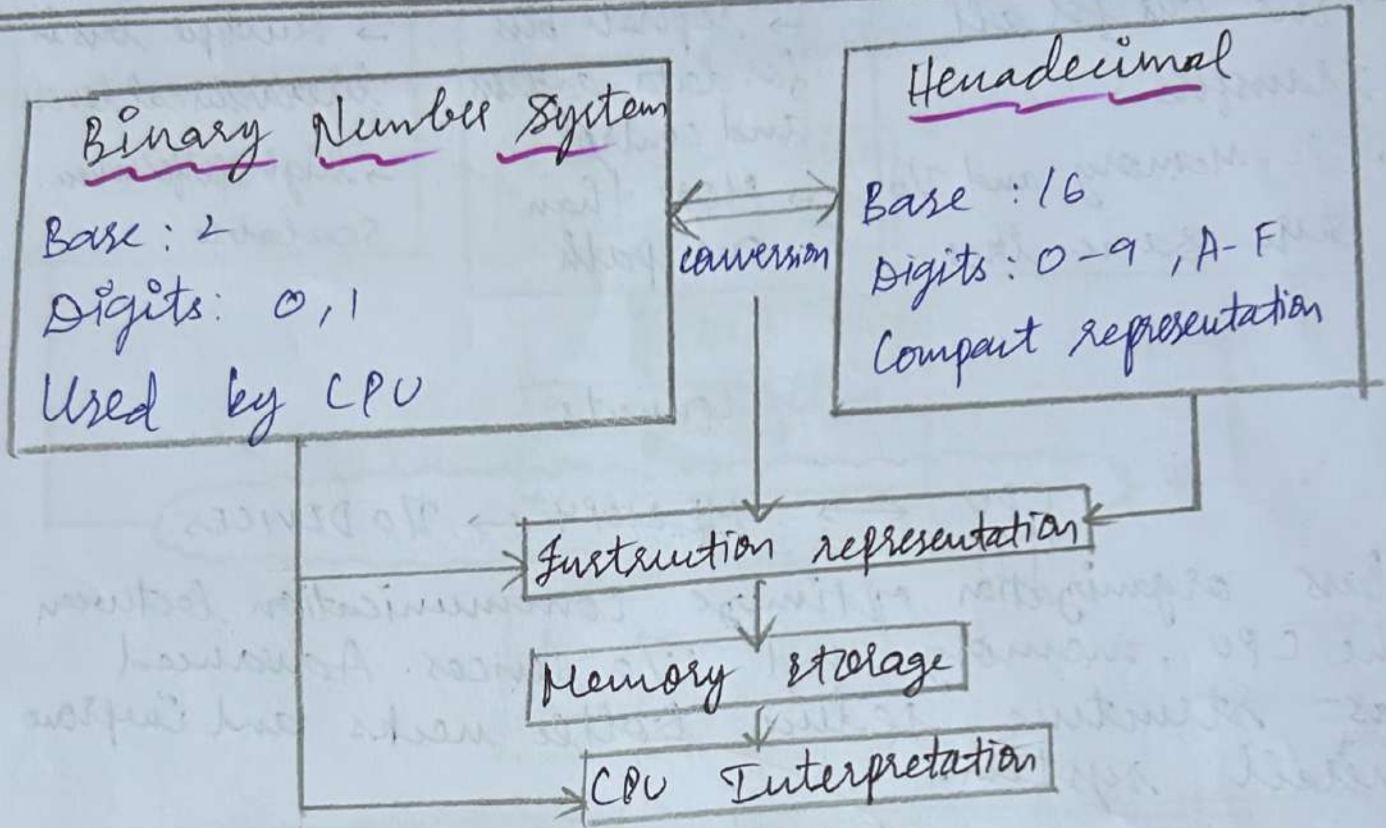
Bus organization optimize communication between the CPU, memory and I/O devices. Advanced bus structure reduce bottle necks and improve overall system.

4. 8085 and 8086 Architecture comparison

8085		8086
8bit Microprocessor	⟷	16 bit microprocessor
64 Kb Memory address	⟷	1MB Memory space
5 General purpose register	⟷	8 General purpose register
Simple addressing mode	⟷	Advanced addressing mode
Lower performance	⟷	Higher performance

compared to 8085 the 8086 executes instruction faster and manages memory more efficiently using 16-bit architecture and memory segmentation

5.



Binary enables the CPU to execute instruction directly, while hexadecimal provides a compact and readable format for memory addresses

Example : Binary: 1010 1111
Hexa : A F